CE/CS/SE 3354 · Software Engineering · Fall 2026

Project Milestone P2

Analysis and High-Level Design

MidMeet

Team	Team 03, section 005
Document version	1.0 (baseline for P2; grows from your P1 version 0.x)
Date	11/8/26
Repository URL	https://github.com/akao335/MidMeet
Git tag for this milestone	p2 (points at the commit that contains this exact PDF)
Submitted on Canvas by	[one member only]

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0 Revision History, Contribution Record and Changes Since P1

All three tables below are graded. The contribution record carries the individual 20% of this milestone and must agree with your repository history.

Version	Date	Author	Summary of change
0.x	10/18/26	Team 3	P1 baseline (copy your last P1 row)
1.0	11/8/26	Team 3	P2 baseline: analysis and high-level design added

0.1 Individual contribution record (P2)

Team member	Sections and artifacts this person produced for P2	Evidence in the repository (commit hashes, PR numbers)	Reviews given (PR numbers)
Alison Kao	4.1, 4.2, 4.3	PR#17(approved by vedasree05)	PR#18 (Vedanshi's section) PR#15
Yair Parkhov	6.1,6.2,6.3,6.4	PR#15	
Nandhana Kamalabaskaran	5.1-5.3		
Vedanshi Sreeramoju	7.1-7.2, 3.4 use case diagram, 3.5 update	Commits 97974c2, 0dcd972; PR #16 (approved by yairp77); PR #18; Issue #9	PR #17 (Alison's Section 4, approved)
Nicholas Jackson	8		
Sameeraa Kandalgaonkar			
Tanay Munnangi	6.5,6.6		

0.2 Changes since the P1 baseline

Change ID	What changed (IDs affected)			Why (P1 feedback, new finding, spike result)	Knock-on effect	Agreed on
CH-01	3.5: P1 matrix moved to 9.1; FR-11 added; note corrected so FR-10 and FR-11 are the intended gaps, not FR-09	P1 feedback: FR-11 missing from the matrix; FR-09 wrongly called WON'T	9.1 now carries all requirement traces	11/8/26		
CH-02						
CH-03						

1 to 3 Introduction, Overall Description and Requirements (carried forward)

1 Introduction

1.1 Purpose of this document and of the system

This document is the baseline requirements specification for MidMeet. It is written for the course instructor and teaching staff, and for the project team, who will use it as the starting point for design and testing. It records what the system must do, the limits it must work within, and the questions we have not yet answered.

MidMeet helps two people who live in different places to choose a fair, real place to meet. The problem belongs to anyone who has to meet a friend or another person somewhere between two homes. Today, people guess the halfway point on a map, and the current online maps do not account for if the driving times between each person are equal. MidMeet takes each person's starting point and suggests real, named places where the two driving times or distance are as close to equal as possible.

1.2 Scope

In scope — we will build this	Out of scope — we will not, and why
Two participants, each with a start location	Three or more participants. This will be harder to define the midpoint, so it is a later release
Balancing travel by distance (miles) or by time (minutes)	User accounts and friend lists: not needed to meet the core goal.
Real, named places with addresses, filtered by type (e.g coffee, restaurant, park)	Booking tables or reservations needs venue integrations and not needed to meet the core goal.
A map showing both starts and the results	Walking, bicycling and public transit: each needs different routing data whose availability we have not confirmed
Driving time and distance for each participant, and the fairness gap, for each place	Live traffic and "leave at" times: depend on provider features we may not have.
A maximum driving limit per participant	

Why the exclusions were excluded:

We limited this release to driving so the team can build and test one routing method in a single semester. Groups, other travel modes, reservations and accounts each add substantial work that does not help answer the core question of where two drivers should meet.

1.3 Definitions, acronyms and abbreviations

Term or abbreviation	Meaning as used in this project
User	The person who operates the system, enters both addresses and picks the place
Participant	One of the two people who will meet. The user may be one of them. The other participant never uses the system
Start location	The address or position a participant will travel from
Driving miles	The road distance, in miles, a participant would drive to a place.
Driving time	The estimated minutes a participant needs to drive to a place, as returned by the routing provider
Midpoint	Two meanings. Geographic: the point halfway along a straight line between the two start locations, which ignores roads. Fair: the place where both participants' trips are closest to equal in the chosen measure. This project finds the fair midpoint and uses the geographic one only as a starting area for the search.
Place	A real, named location with a street address, not just coordinates
Geocoding	Converting an address to coordinates
Search radius	The area around the midpoint in which candidates places are looked up
Provider	The third-party service that supplies map, geocoding, routing and place data
Fairness gap	The difference between the two participants, driving times (minutes) or driving distances (miles), whichever the user chose. A smaller gap means a fairer place

1.4 References

[R1] Google Maps Platform Documentation https://developers.google.com/maps/documentation

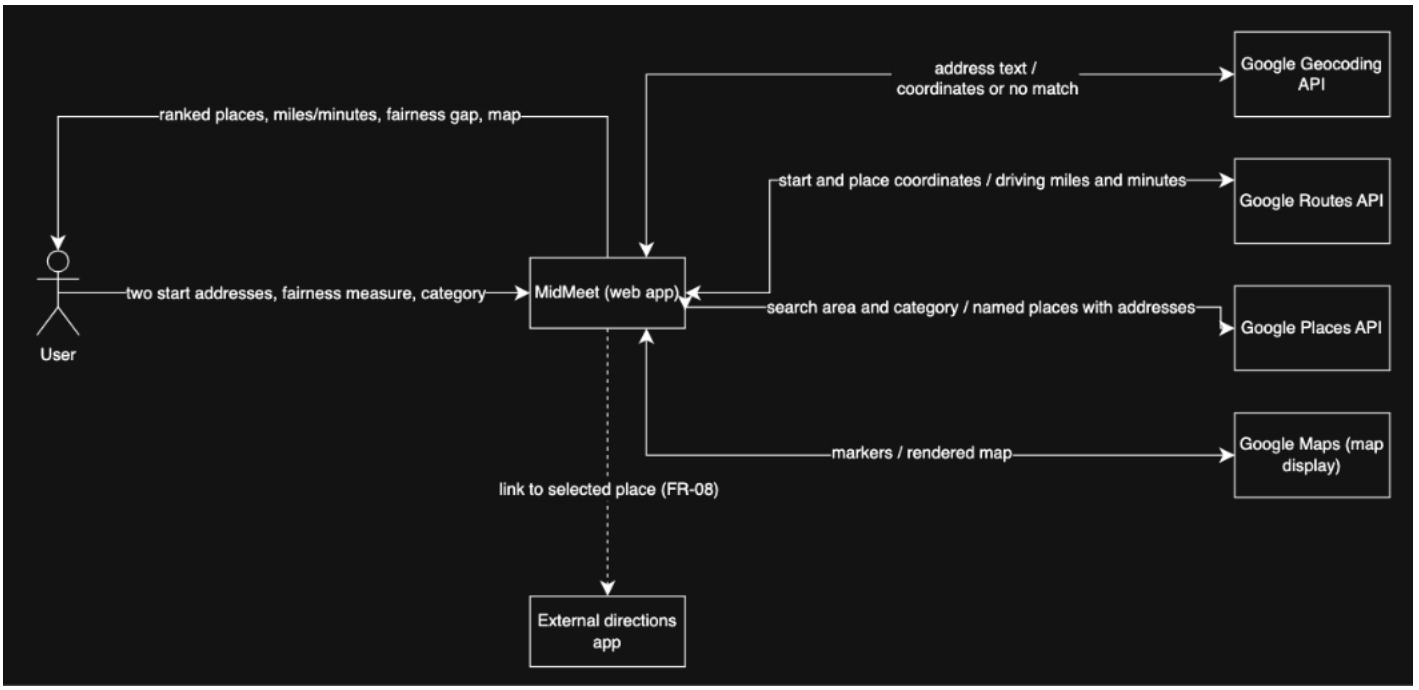
[R2] Meeting Spaces in Dallas https://www.visitdallas.com/blog/coworking-spaces-in-dallas/

[R3] Three ways to add map implementation https://mapsplatform.google.com/resources/blog/three-ways-add-map-implementation-your-app-and-when-use-each/?utm_experiment=13103221

[R4] Student Interview https://github.com/akao335/MidMeet/blob/main/docs/Student%20Interview

2 Overall Description

2.1 Product perspective and context



Everything inside the MidMeet box is what our team builds: taking the two start addresses and the user's choices, finding the search area, calculating each participant's driving miles and minutes and the fairness gap, ranking the places, and displaying the results. Map, geocoding, routing and place data come from Google Maps Platform (R1) and sit outside the boundary because we cannot produce this data ourselves in one semester (CON-01). The external directions app is also outside, because MidMeet only links to it (FR-09) and does not give directions itself. The other participant is not on the diagram, because they never use the system (2.2) and only see what the User shares with them. User accounts, reservations and other travel modes are out of scope (1.2), so no account or booking system appears.

2.2 Stakeholders and how you reached them

Stakeholder	Role in relation to the system	What they need from it	How you learned this
People meeting up (friends, family, partners)	Primary user; enters both addresses and picks the place	Get a fair, real place to meet in a couple of minutes, with each person's drive time shown so they can justify the choice	Assumed, based on the team's own experience arranging meetups.
Other participant (friend, family, partner)	Never uses the system; their address is entered by the organizer	A result that is actually fair to them, and their address not stored or shared	Assumed. This stakeholder exists because the summary

			has one person entering both addresses (FR-01)
Course instructor and TAs	Never use the system; evaluate it and set the rules it is built under	Testable, traceable requirements; repository access; delivery by the course deadlines	P1 template and course syllabus
Project team	Builds, operates and pays for the system for the semester	Stay within free API usage; scope that seven people can finish in one semester	Team decision,
Map, routing and places data provider	Never uses the system; supplies all location data	Compliance with its terms: attribution, usage limits, limits on storing its data	Provider terms of service
Venues shown in results (cafés, restaurants, parks)	Never use the system; appear as recommendations	Correct name, address and category so they aren't misrepresented	Assumed

2.3 User characteristics

Organizer. Most organizers are students or young adults on a phone, often while messaging the other person, and want an answer within a couple of minutes. They know both people's general locations but may not know the exact address. They might type a partial address, a neighborhood, or a landmark, so the system must handle near-misses (FR-02). They can't be assumed to know map terms, and they can't be assumed to know the difference between balancing by miles and by minutes, or what a "fairness gap" is. Those choices need plain-language labels. The two participants' markers on the map must be distinguishable by more than color alone, for color-blind users.

Other participant. This person never touches the system, but they constrain it in two ways. Their address is entered by someone else without them being present, which is why start locations must not be kept (QR-02). The result also must be easy for the organizer to screenshot or share, because the other participant sees only that.

2.4 Constraints

#	Constraint	Source
C1	Map, geocoding, routing and place data must come from a third-party provider, because we cannot build this data ourselves	Team capacity in one semester; carried forward as CON-01
C2	The system must run at no cost to the team, using only providers' free usage tiers	Team has no project budget and the course provides none CON -02

C3	Use of provider data must follow the provider's terms, including attribution and limits on caching or storing results	Provider terms of service CON -03
C4	The release must be delivered by the course milestones (P2 on 8 Nov, P3 (Dec 9th)	Course syllabus
C5	Source code and project history must be kept in the team GitHub repository, with access for the instructor	Course requirement, P1 template 4.4

2.5 Assumptions and dependencies

#	Assumption or dependency	What breaks if it does not hold
A1	Access to Routes, Geocoding, and Places API from Google or potentially a different source	The ability to produce optional meeting points, routes, and time estimations within the time constraints of the project.
A2	Google's estimated driving times are close enough to real driving times that equal estimates mean a fair meetup	A place we show as fair could give one person a much longer drive causing our main goal to fail. QR04 could pass in testing but real users could get unfair results
A3	Users enter addresses the geocoding service can find, such as a full street address, zip code, or a well-known place name.	FR 01 fails often and users keep running into the FR2 error. We would have to rely on address autocomplete, which is currently outside of our scope.
A4	At least 3 places of the chosen category exist within the starting search radius around the midpoint for most address pairs in the DFW area.	FR04 isn’t met, because the system can’t show 3-10 places. In sparse areas, searches would end up in the “too few places” case which Q2 and Q3 deals with more often than they succeed.

2.6 Feasibility summary

Feasibility gate	Verdict	Evidence behind the verdict
Technical	Feasible, pending a spike	The core task is geocoding two addresses, searching places near the midpoint, and getting drive times from 2 origins to each candidate. All of these are standard API calls. Few of the team members have used React and are open to exploring the Google Maps API before. Spike: a script that geocodes two addresses and returns drive times to 10 nearby cafés.
Operational	Feasible	No accounts, no stored data, no staff. It runs as a web page the team deploys on a free host Vercel or Netlify. Users need only a browser and two addresses.

Schedule	Feasible with the reduced scope	The scope was cut to 2 people, driving only, and no accounts (1.2), so it fits the course's timeline. The milestones in 4.2 have named owners. Main risk: API integration taking longer than planned (see 4.3).
Economic / resource	Feasible at $0	One search uses 2 Geocoding calls, 1 Places Nearby Search, and a 2 × 20 Route Matrix (40 elements). Google Maps Platform gives 10,000 free events/month per Essentials SKU and 5,000 per Pro SKU. Route Matrix is the binding limit at about 250 searches/month, more than the ~200 needed for testing (QR-01) and demos (pricing page, retrieved 18 Oct 2026). Fallback: OpenStreetMap services (Nominatim, OSRM, Overpass) have no fees.

3 Specific Requirements

3.1 Functional requirements

ID	Requirement — one “shall” sentence	Priority	Source	Acceptance criterion
FR-01	The system shall accept a start location from each participant entered as an address.	MUST	S1, S2, R1	A tester can confirm this by entering two addresses and see both marked
FR-02	The system shall display an error message beside an invalid address field while preserving the previously entered data.	MUST	S1, R1	When a tester enters an unacceptable text as an address, the message appears next to the invalid field and the valid address is unchanged.
FR-03	The system shall let the user choose whether the fairness gap is measured in miles or minutes, defaulting to driving minutes.	MUST	S1, R1, R4	A tester runs each of 10 test address pairs twice, once selecting miles and once minutes. With miles selected, the gap is labelled in miles, and the list is sorted by ascending mile gap. With minutes selected, the gap is labelled in minutes, and the list is sorted by ascending minute gap.
FR-04	The system shall present between 3 and 10 real, named places with street addresses as candidate meeting spots.	MUST	S1, S6, R1, R2, R4	Across 10 test address pairs, every result list has 3-10 places, each with a name and address.

FR-05	The system shall let the user filter candidate places by category	MUST	S1, R1, R4	A tester selects a category, and every returned place is within that category. For example, when a tester selects "restaurant" all places in the result list are restaurants.
FR-06	The system shall display, for every candidate place, both participants' driving miles, driving minutes, and the fairness gap in the user's chosen measure.	MUST	S1, S2, R1, R4	For the top 3 results of 10 test address pairs (30 rows), each row shows two distances in miles and two times in minutes that match the routing provider's values after rounding, and a gap equal to their difference in the chosen measure, labelled with its unit.
FR-07	The system shall display both start locations and all candidate places on a map	MUST	S1, R1, R4	A tester confirms every listed place, and both starts have a marker
FR-08	The system shall display a message when no candidate meeting places are found	MUST	S1	A tester searches an area or applies a category filter with no matching places and confirms that a message appears indicating no places were found.
FR-09	The system shall provide a link from a selected place to an external directions app	COULD	S1, R1, R4	A tester clicks the link, and the directions app opens with that place.
FR-10	The system shall support three or more participants	WON'T	S4	Agreed out of this release (see 1.2) A tester confirms that the interface offers no way to enter a third address.
FR-11	The system shall support multiple transportation modes	WON'T	S4	Agreed out of this release, we will focus on car transportation for this release (see 1.2) A tester confirms that the interface offers no travel-mode option and all times and distances are for driving

3.2 Quality requirements

ID	Requirement — with threshold, condition and method	Priority	Source	How it will be verified
QR-01	95% of searches shall return results within 5 seconds, measured over 100 test searches on a normal mobile connection.	MUST	S1, R1, R4	A script times 100 searches and reports the 95th percentile
QR-02	Zero entered addresses shall be retained after the session ends, verified by inspecting stored data after 10 completed sessions.	MUST	S2, R1	Inspect storage after each session and inspect outgoing network requests
QR-03	At least 4 of 5 first-time testers shall complete a search, from entering two addresses to choosing a place, within 2 minutes without help	SHOULD	S1	A timed, moderated usability session with 5 testers who haven't seen the system
QR-04	For at least 18 of 20 predefined test address pairs, the top-ranked place's fairness gap shall be no more than 20% of the longer participant's driving time.	MUST	S1, S2	Compare the top result for each of the 20 pairs against the routing provider's values.

3.3 Constraint requirements

ID	Constraint — one sentence	Source	Why it cannot be negotiated
CON-01	The system shall obtain map, routing and place data from third-party providers	C1, R1	We cannot produce this data ourselves in one semester
CON-02	The system shall operate at no cost to the team, using only third-party providers' free usage tiers	C2, R1	The team has no project budget, and the course provides none.
CON-03	The system shall use provider data only as the provider's terms permit, including attribution and limits on caching or storing results	C3, R2	Breaking the terms can end the team's access to all location data
CON-04	The project shall be delivered by the course milestones, with all source and history kept in the team GitHub repository, accessible to the instructor	C4, C5	The course sets these dates and the repository rules.

3.4 Initial use cases

MidMeet has four use cases, each started by the User to reach a goal of their own. Entering addresses, choosing a category, and viewing the map are steps inside UC-01, not separate use cases. UC-03 and UC-04 extend UC-02, because they are optional and only happen after a place has been chosen. UC-01 to UC-03 are expanded turn by turn in Section 4.

- UC-01 Find fair meeting places (User)
- UC-02 Compare and select a place (User)
- UC-03 Get directions to the chosen place (User; extends UC-02)
- UC-04 Share the chosen place (User; extends UC-02)

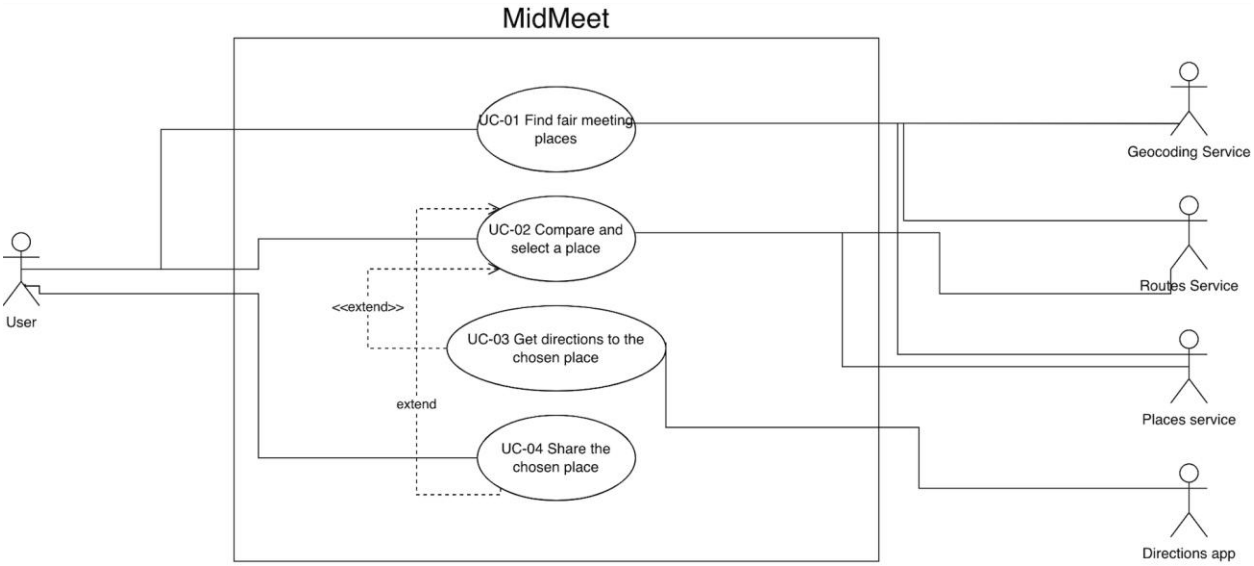


Figure 3.4 Use case diagram

3.5 Requirements to use case traceability

The requirements-to-use-case matrix is carried into the extended traceability matrix in Section 9.1, which adds FR-11 and corrects the earlier note that called FR-09 a WON'T item.

4 Use-Case Analysis

Chapter 7, first half. Here you take the use cases from section 3.4 and write the most important of them out turn by turn, still with the technology switched off.

4.1 Use cases selected for analysis

Use case	Name	Why it was selected	Non-trivial steps
UC-1	Find Fair meeting places	Carries the core promise of the product and touches most of our requirements (FR-01 to FR-08, QR-01, QR-02, QR-04). It depends on outside data and has four failures the User can see: an address that cannot be found, two addresses at one location, too few places, and an unavailable provider.	(10): locating both addresses, finding places near the middle, getting both participants' travel figures, ranking the places and showing them. Steps (1) to (9) only record or acknowledge the User's input.
UC-2	Compare and select a place	This is where fairness is judged. Changing the category or the measure re-ranks the places, and QR-04 holds only if the ranking is right in both measures.	(2): finding and ranking places for a new category. (4): re-sorting by the other measure. Step (6) only shows details already held, so it is trivial.
UC-3	Get directions to the chosen place	The only use case that leaves MidMeet for an external system, and the hand-over can fail in a way the User sees. We are least sure how this hand-over behaves.	(2): handing the chosen place to the directions app.

4.2 Expanded use cases

Copy the block below once for each use case in 4.1.

UC-01 Find Fair Meeting Places

Primary actor	User
Supporting actors and external systems	Location data provider(supplies address lookup, place data and travel figures; outside MidMeet, R1)
TUCBW (this use case begins when)	The User asks MidMeet for a fair place for two participants to meet
TUCEW (this use case ends when)	The User sees the ranked places, each with both participants' travel figures, on a list and on a map.

Preconditions	MidMeet is open on the User's device.
Postconditions	The User has seen 3 to 10 real, named places ranked by fairness gap, each with both participants' driving miles and minutes. Both start locations and all listed places are on the map. No start address is kept after the session ends (QR-02).
Traces to requirements	FR-01, FR-02, FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, QR-01, QR-02, QR-03, QR-04, CON-01

Actor input / action	System response
	(0) The system shows an entry for each participant's start address, a choice of fairness measure with minutes selected, a choice of place category, and no results
(1) User enters the first participant's start address	(2) The system shows the address as entered, beside participant 1
(3) The User enters the second participant's start address.	(4) The system shows the address as entered, beside participant 2.
(5) The User chooses to measure fairness in miles or in minutes.	(6) The system shows the chosen measure as selected.
(7) The User chooses a place category, such as coffee, restaurant or park	(8) The system shows the chosen category as selected
(9) The User asks for results	(10) The system shows both start locations and 3 to 10 places of the chosen category on a map and in a list ranked by smallest fairness gap. Each place shows its name, street address, both participants' driving miles and minutes, and the fairness gap in the chosen measure, labelled with its unit.

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A1	10	A start address is empty or cannot be found	The system shows an error beside that address and keeps everything else the user entered (FR-02). The User correct the address	Rejoins at step 1 or 3
A2	10	Both addresses refer to the same location	The system tells the User and asks them to check the addresses. Entries are kept	Rejoins at step 1

Alternate flow	At step	Condition	What happens	Ends or rejoins at
A3	10	Fewer than 3 places of the chosen category are found, even after search area is widened	The system tells the User too few places were found, stating that none were found when there are none (FR-08), and keeps all entries. The User may choose another category	Rejoins at step 7
A4	10	The provider does not answer in time or fails	The system shows a "try again" message and keeps everything the User entered.	Rejoins at step 9

UC-02 Compare and Select a Place

Primary actor	User
Supporting actors and external systems	Location data provider (supplies place data for a new category; outside MidMeet, R1)
TUCBW (this use case begins when)	The User, looking at the ranked places, wants to compare the options and settle on one.
TUCEW (this use case ends when)	The User sees the selected place highlighted on the map with its details.
Preconditions	UC-01 has finished and a ranked list of places is on screen
Postconditions	One place is selected. The User can see its name, street address, both participants' driving miles and minutes, and the fairness gap.
Traces to requirements	FR-03, FR-04, FR-05, FR-06, FR-07, FR-08, QR-03, QR-04, CON-01

Actor input / action	System response
	(0) The system shows the ranked places in a list and on the map with both start locations, and the category and fairness measure currently in use.
(1) The User changes the place category, for example from coffee to park	(2) The system shows 3 to 10 places of the new category, ranked by smallest fairness gap in the current measure, on the list and on the map
(3) The User switches the fairness measure between minutes and miles	(4) The system shows every place with its gap in the new measure and unit, sorted by smallest gap
(5) The User selects one place	(6) The system highlights that place on the map and shows its name, street address, both participants' driving miles and minutes, and the fairness gap.

Alternate flow	At step	Condition	What happens	Ends or rejoins at

A1	2	No place of new category exists near the middle	The system tells the User and keeps the previous category and results on screen	Rejoins at step 1
A2	2	The provider does not answer in time or fails	The system shows a "try again" message and keeps the previous results on screen.	Rejoins at step 1
A3	4	The fairest place in the new measure is not the fairest in the old one.	The system shows the new order and tells the User the order changed because the measure changed.	Rejoins at step 5
A4	6	The User wants directions to the selected place	The system offers directions; UC-03 begins.	Ends here; UC-03 begins.
A5	6	The User wants to send the result to the other participant	UC-04 (Share the chosen place) begins.	Ends here; UC-04 begins

UC-03 Get Directions to the Chosen Place

Primary actor	User
Supporting actors and external systems	Directions app (a separate application on the User's device that shows routes; outside MidMeet)
TUCBW (this use case begins when)	The User, having selected a place, asks for directions to it.
TUCEW (this use case ends when)	The User sees the directions app showing the selected place as the destination.
Preconditions	UC-02 has finished: one place is selected and its details are on screen.
Postconditions	The directions app is showing the selected place's name and address as the destination. The selected place is still selected in MidMeet.
Traces to requirements	FR-09

Actor input / action	System response
	(0) The system shows the selected place's name, street address and details, with the option to get direction to it
(1) The User asks for directions to the selected place.	(2) The system opens the direction's app with the selected place as the destination, showing its name and address

Alternate flow	At step	Condition	What happens	Ends or rejoins at

A1	2	The directions app cannot be opened or does not accept the place	The system tells the User that directions could not be opened and shows the place's full street address so it can be copied. The selected place stays selected.	Ends; the User may rejoin UC-02 at (5).

4.3 Self-review log

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-01	Step 0 present	P1 had no starting screen, so nothing said what the User sees before acting	Added (0): both address entries, the measure choice with minutes selected, the category choice and no results.
UC-01	One turn per row	P1 steps 1 to 4 were all User input with no system response between them, and steps 5 to 7 were system work the User only waits for.	Split the input into separate turns, each with a response, and merged the waiting steps into response (10).
UC-01	Ends with a perceivable acknowledgment	P1 steps 2 and 3 (choosing the measure and category) were never acknowledged.	Added responses (6) and (8) that show the choice as selected.
UC –01	Precondition or alternate flow	P1 listed "internet connection and provider services reachable" as a precondition, but a provider failure can happen while MidMeet works normally. It is already flow 6b.	Reduced the precondition to "MidMeet is open on the User's device" and kept the failure as A4.
UC-01, UC02	No technology	P1 named "Geocoding service, Routes service, Places service (Google Maps Platform)" and said the system "geocodes" addresses.	Replaced them with "Location data provider" and "locates". The provider is named only in the references.

Use case	Check from the Lesson 13 review checklist	Defect found	How it was fixed
UC-02	Real alternate flows	P1 left open whether the User is told the order changed when the measure changes (its flow 5a). P1 also gave no failure for a category change.	Decided that the system tells the User (A3), and added A1 and A2 for the category change. A switch of measure needs no new provider request, because both miles and minutes are already held from UC-01.
UC-03	Step 0 and acknowledgment	P1 gave UC-03 no turn-by-turn form, so the hand-over to the directions app had no visible end.	Wrote (0) to (2), ending when the directions app shows the destination, and added A1 for a hand-over that fails.
All	Actor items are in the domain model	The actor columns name these items: a start address for each participant, the fairness measure, the place category, the selected place, and the directions request.	These become required attributes in the domain model (5.1), and check K1 in 5.3 confirms them.

5 Analysis Models

Chapter 7, second half. What has to exist inside the system for the use cases in section 4 to happen, and how those parts talk to each other.

5.1 Domain model

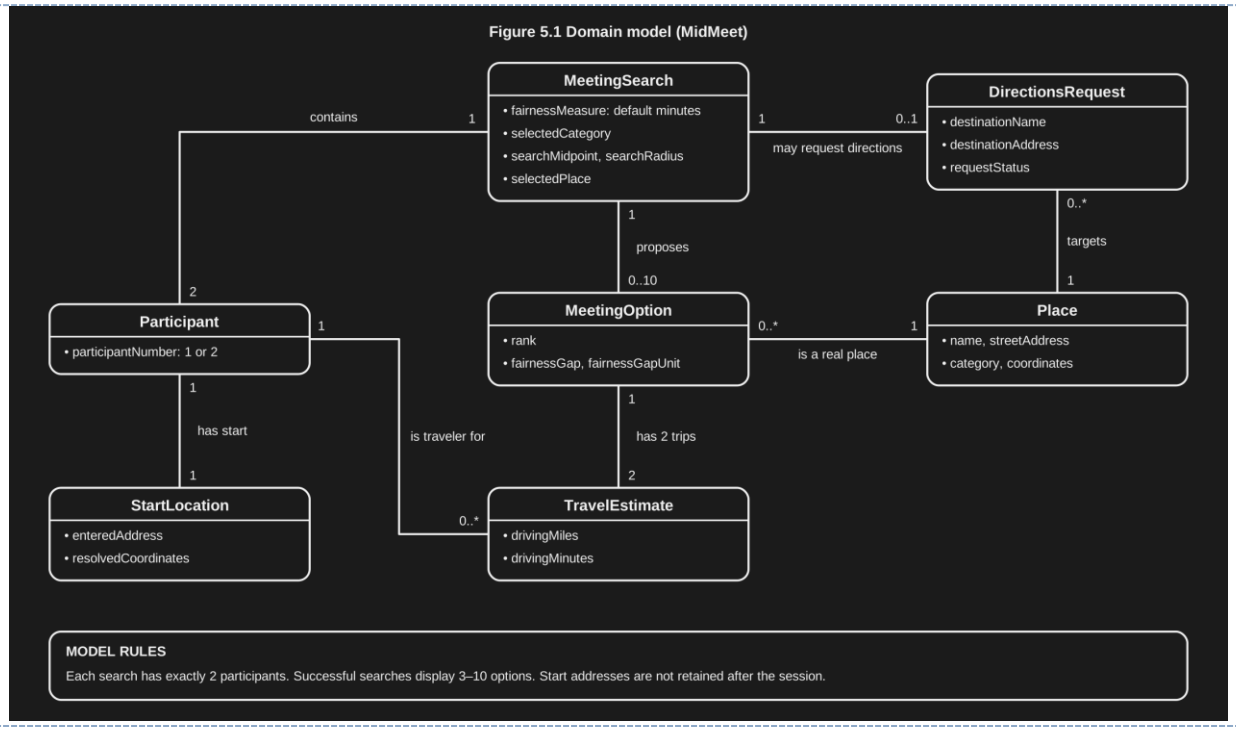


Figure 5.1 Domain model (conceptual classes, attributes, associations with multiplicities)

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
MeetingSearch	A search for fair meeting places between two participants	fairnessMeasure, selectedCategory, searchMidpoint searchRadius, selectedPlace,	2 Participants, 0..10 MeetingOptions, 0..1 DirectionsRequest
Participant	One of the two people meeting	participantNumber	1 MeetingSearch, 1 StartLocation, 0...* TravelEstimates
StartLocation	Starting address of a participant	enteredAddress, resolvedCoordinates	1 Participant
MeetingOption	A candidate meeting place ranked by fairness	rank, fairnessGap, fairnessGapUnit,	1 MeetingSearch, 1 Place, 2 TravelEstimates
Place	A real meeting destination	name, streetAddress, category, coordinates	0...* MeetingOptions, 0...* DirectionsRequests

Class	What it represents (one line)	Key attributes	Associations (with multiplicity)
TravelEstimate	Estimated driving trip from one participant to a meeting option	drivingMiles, drivingMinutes	1 Participant, 1 MeetingOption
DirectionsRequest	Request to open external directions	destinationName, destinationAddress, requestStatus	1 MeetingSearch, 1 Place

5.2 Object interaction models

Copy the block below for at least two of the use cases in 4.1; preferably including your riskiest.

5.2.1 UC-01 Find Fair Meeting Places

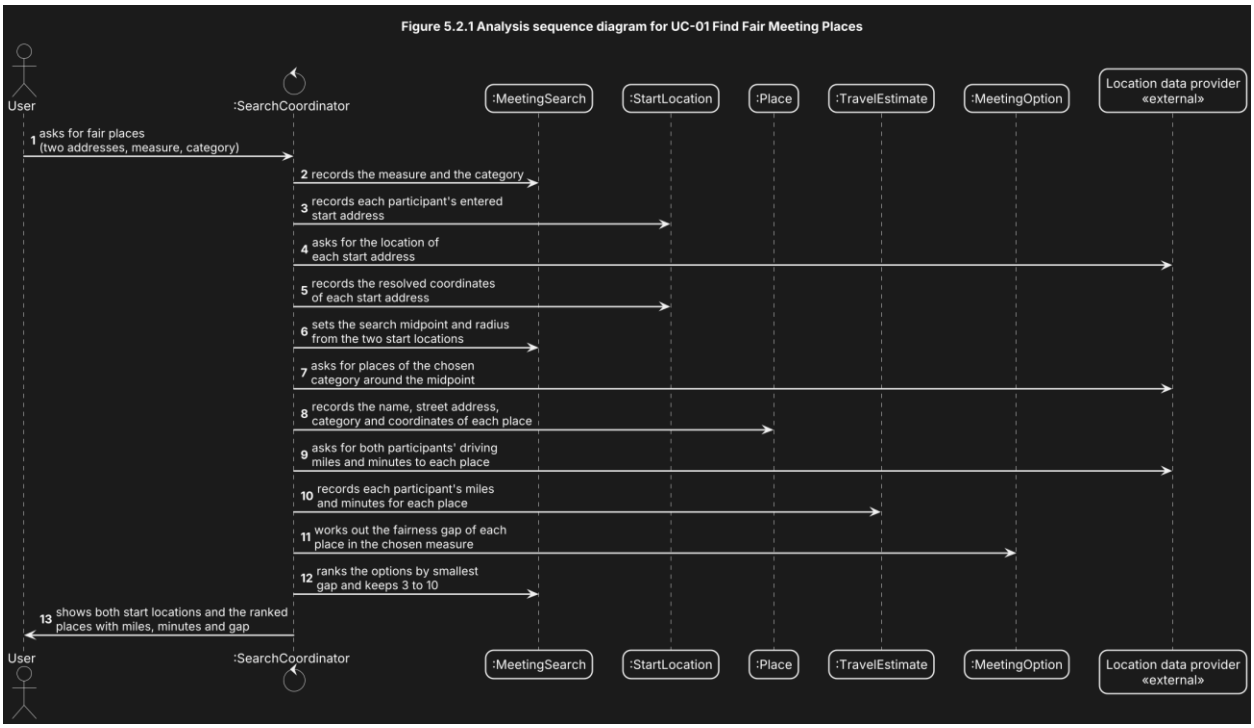
(a) Non-trivial steps. List the steps from 4.2 that need objects inside the system, and say in one line why each of the others is trivial.

Step 10 is non-trivial because the system must resolve both starting addresses, calculate the midpoint, retrieve nearby places, obtain driving estimates for both participants, calculate fairness gaps, and rank the candidate places. Steps 1–9 are trivial because they involve entering information, selecting preferences, and acknowledging the user's choices.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	User	Requests meeting places	Addresses, category	SearchCoordinator
2	SearchCoordinator	Records search details	FairnessMeasure, selectedCategory	MeetingSearch
3	SearchCoordinator	Resolves address	enteredAddress	StartLocation
4	SearchCoordinaotr	Finds nearby places	SearchRadius, category	Location Data provider
5	SearchCoordinaotr	Gest driving estimates	Partcipant, place	Start Location
6	SearchCoordinaotr	Calculates and ranks options	Travelestimae, fairness gap	MeetingSearch
7	SearchCoordinaotr	Displays ranked results	Meetingoption, start locaiton	Location data provider

(c) Analysis sequence diagram.



5.2.2 UC-02 Compare and Select a Place

(a) Non-trivial steps. List the steps from 4.2 that need objects inside the system, and say in one line why each of the others is trivial.

Steps 2 and 4 are non-trivial. Step 2 requires retrieving places in a new category, obtaining updated travel estimates, and ranking the new options. Step 4 requires recalculating fairness gaps using the new measure and reordering the options. Steps 1, 3, and 5 are trivial user actions, while step 6 displays the selected place's existing information.

(b) Scenario table.

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
1	User	Changes category	selectedCategory	SearchCoordinator
2	SearchCoordinator	Finds new places	Place, searchRadius	Location data provider
3	SearchCoordinator	Calculates and ranks options	TravelEstimate, fairnessGap	MeetingOption
4	User	Changes fairness measure	fairnessMeasure	SearchCoordinator
5	SearchCoordinator	Recalculates and sorts results	TravelEstimate, fairnessGap	MeetingOption

Step	Subject (who acts)	Subject action (what it does)	Other data / objects (what is used)	Object acted upon (who receives)
6	User	Selects a place	selectedPlace	SearchCoordinator
7	SearchCoordinator	Records selection and displays details	Place, MeetingOption	User

(c) Analysis sequence diagram.

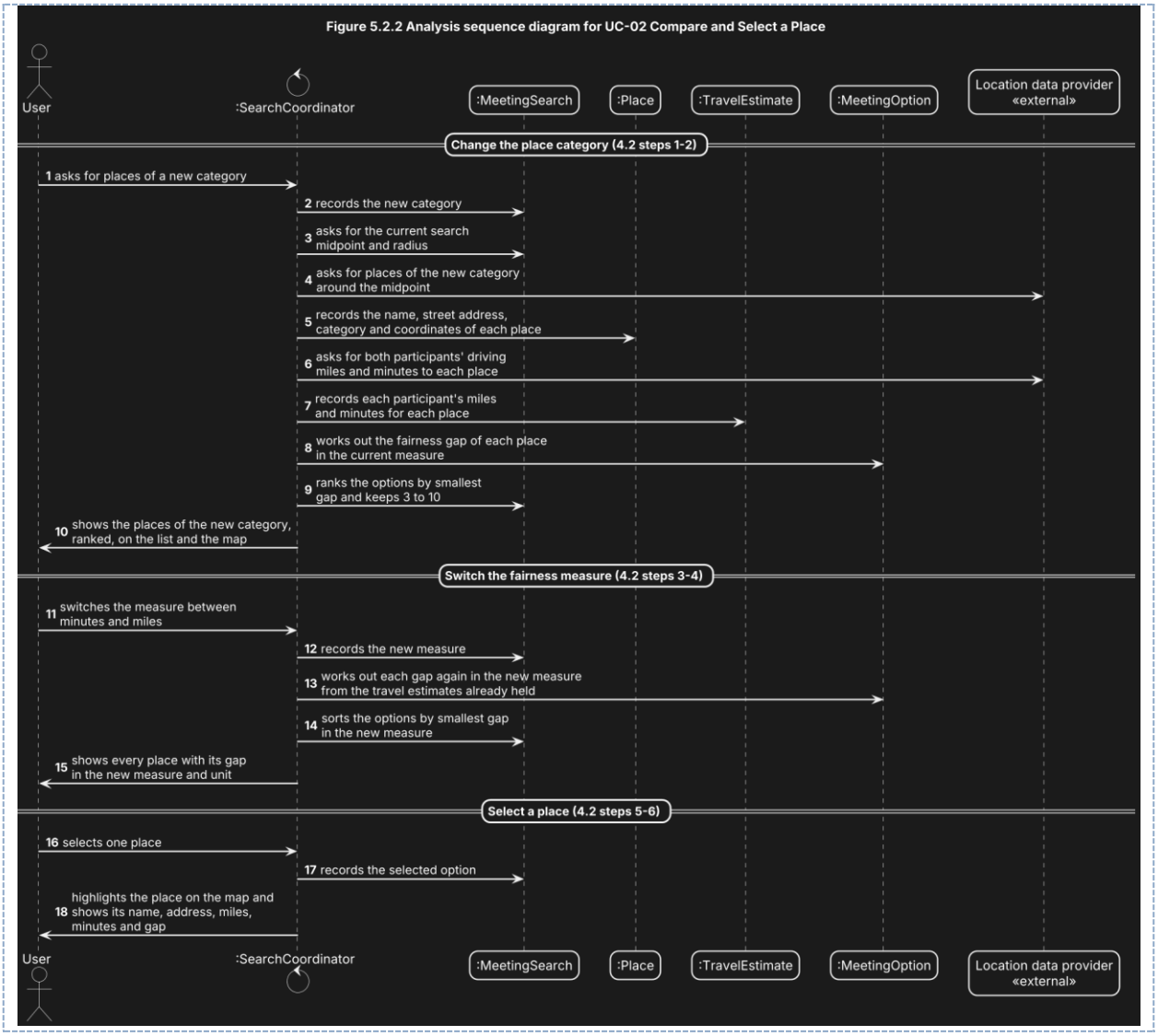


Figure 5.2.2 Analysis sequence diagram for UC-02; message numbers match the scenario table

5.3 Consistency cross-check

Check	Result (pass, or what failed)	Fixed in
K1 Every item in an actor column (4.2) is an attribute in the domain model (5.1)	Pass, Addresses, fairness measure, category, selected place, and directions request are represented.	5.1
K2 Every object on a sequence diagram is a domain class or a declared interaction object	Pass, the domain objects are MeetingSearch, startLocation, Place, TravelEstimate, and MeetingOption, SearchCoordinator and Location data provider are interaction objects	5.2
K3 Every message on a sequence diagram corresponds to exactly one scenario-table row	Pass, UC-01 has 13 matching messages and UC-02 has 18 matching messages	5.2.1, 5.2.2
K4 Every non-trivial step in 5.2 (a) has at least one scenario-table row	Pass, UC-01 step 10 and UC-02 steps 2 and 4 are represented.	5.2
K5 Every analyzed use case traces to at least one functional requirement	Pass, UC-01 traces to FR-01-FR-08, and UC-02 traces to FR-03-FR-08.	4.2, 5.2
K6 The last messages of each diagram make the use case postcondition true	Pass, UC-01 ends with ranked places shown. UC-02 ends with the selected place highlighted and its details displayed	5.2

6 High-Level Design (Architecture)

Chapter 8. The rule now reverses: a high-level design that does not say where things run, what they are built from and what that choice cost has not been done.

6.1 Architectural drivers: quality-attribute scenarios

QAS	Quality attribute	Stimulus	Artifact	Response	Response measure	From
QAS-01	Performance	User on a normal mobile connection submits a search.	Search path: web client, Search API, provider calls.	Ranked places appear in the list and on the map.	95th percentile <= 5 s over 100 timed searches.	QR-01
QAS-02	Security	The session ends. The tester checks everywhere the addresses could remain.	Browser storage, server logs and runtime.	Addresses live only in memory. Never stored, logged or put in a URL.	0 addresses found after 10 sessions.	QR-02
QAS-03	Availability	Provider call fails or takes longer than 4 seconds.	Google Connector, Search Controller	A "try again" message. All inputs kept. no partial ranking.	Message within 6 s in 20 of 20 tests where Google fails 100% of inputs kept.	UC-01 A4
QAS-04	Modifiability	The team must switch location data to OpenStreetMap	Google Connector	Only the Google Connector changes. LocationData stays the same.	1 component changed, 0 others.	CON-01, CON-02, A1
QAS-05	Reliability	A tester runs the 20 predefined DFW address pairs.	Fairness Ranker	Places ranked by the gap between real road drive times, fairest first.	Top gap <= 20% of the longer drive for >= 18 of 20 pairs.	QR-04

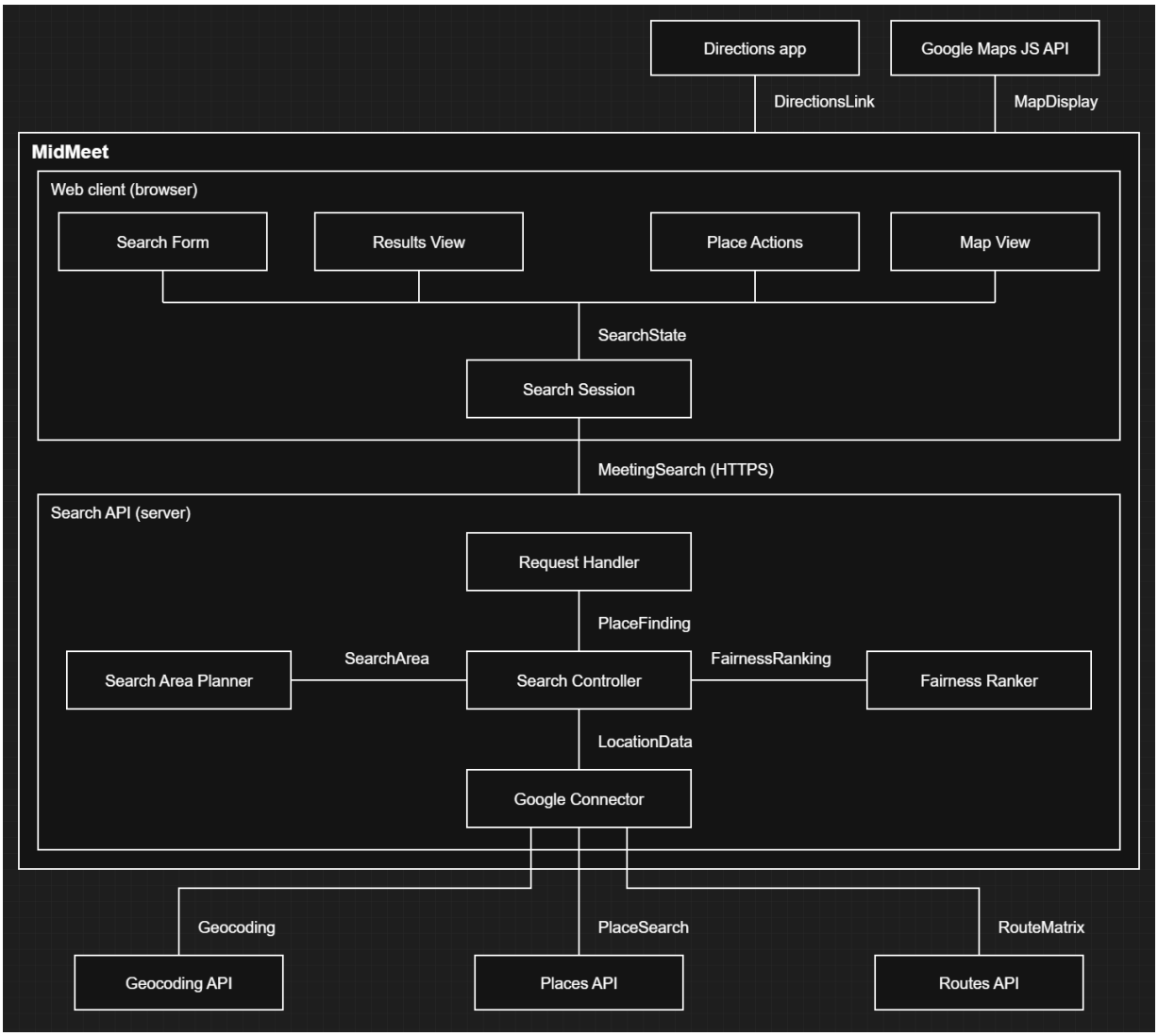
CON-01: All map data comes from Google - only the Google Connector talks to it. CON-02: we use only free plans, so the key stays on our server. CON-03: we don’t save Google’s results.

6.2 Architectural style and alternatives

Style considered	Drivers it serves	What it costs	Verdict
Client-server: the browser is the client; our server holds the logic and talks to Google.	QAS-02, CON-02: the key and addresses stay on our server, which saves nothing. QAS-04: only one part talks to Google.	One extra step per search. We must build the server.	Chosen
No server, calls google directly	Simpler and a bit faster (QAS-01)	The Google key is exposed in every browser (CON-02)	Rejected
Layered with our own database.	Faster repeat searches (QAS-01)	It saves addresses, which breaks QR-02	Rejected

We picked client-server because it keeps our Google key safe (CON-02) and saves no addresses (QAS-02). It adds one step per search, but still fits in 5 seconds (QAS-01).

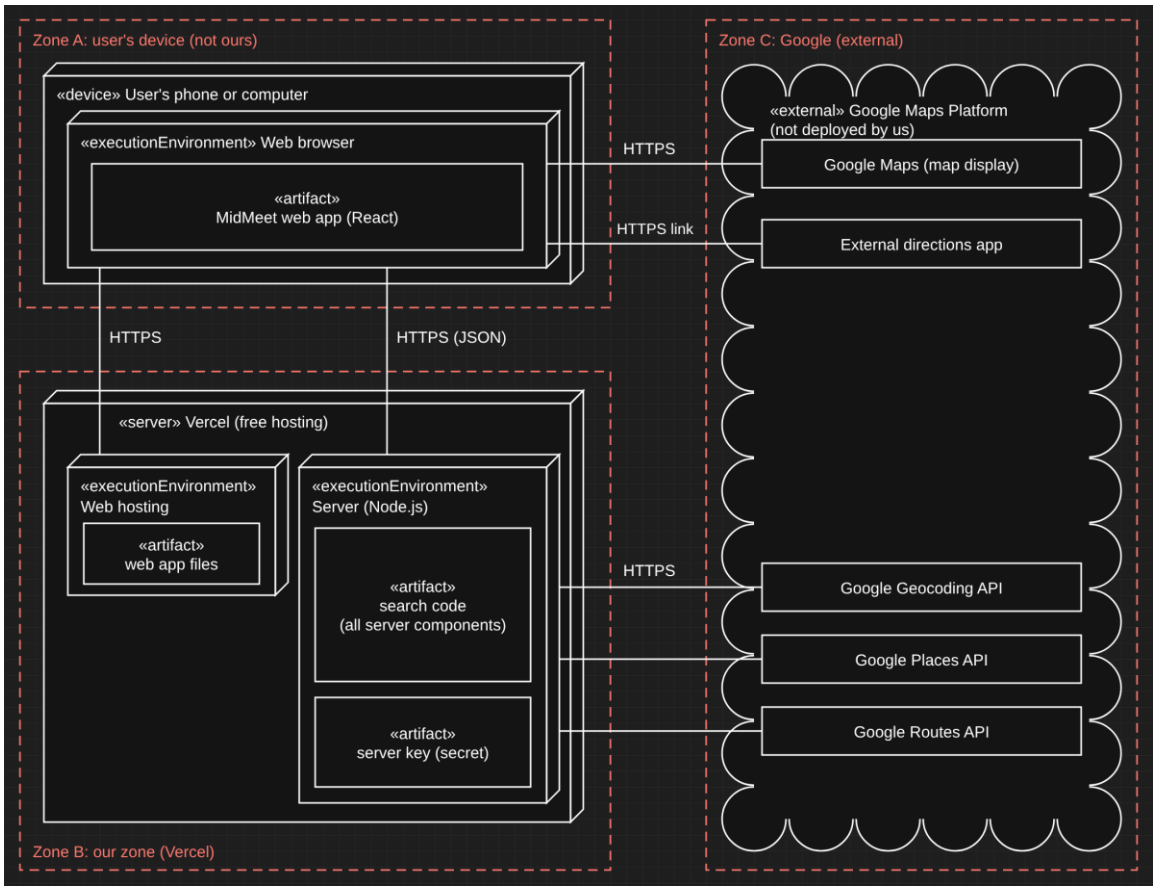
6.3 Component view



Component	Responsibility (one sentence)	Provides	Requires	Answers to (UC / QAS / ADR)
Search Form	Get 2 addresses: the measure and the category, error if address not found.	none	SearchState	UC-01
Results View	Shows up to 10 places with miles, minutes and the gap.	none	SearchState	UC-01 UC-02
Map View	Shows both start points and all places on the map.	none	SearchState MapDisplay	UC-01 UC-02

Place Actions	Makes the directions link and the message for the chosen place	none	SearchState DirectionsLink	UC-03 UC-04
Search Session	Keeps the search in memory only, sends it to the server	SearchState	MeetingSearch	UC-01 UC-02 QAS-01 QAS-02
Request Handler	Checking search request and send back the results	MeetingSearch	PlaceFinding	UC-01 UC-02 QAS-03
Search Controller	Runs the search and makes the search radius bigger if fewer than 3 places are found.	PlaceFinding	SearchArea FairnessRanking LocationData	UC-01 UC-02 QAS-01 QAS-03
Search Area Planner	Finds the midpoint and the search area	SearchArea	none	UC-01
Fairness Ranker	Find the gap for each place and puts the fairest first	FairnessRanking	none	UC-01 UC-02 QAS-05
Google Connector	Talk to google with addresses, places and drive times	LocationData	Geocoding PlaceSearch RouteMatrix	QAS-03 QAS-04 CON-01

6.4 Deployment view



Node	What runs on it (artifacts, components)	Communicates with (protocol)	Trust zone
User's phone or computer	MidMeet web app (React): Search Form, Results View, Map View, Place Actions, Search Session	Web hosting: HTTPS. Server: HTTPS. Google Maps: HTTPS. External directions app: HTTPS link	Zone A: User device- We don't trust input from here
Vercel: Web hosting	Web app files	Web browser: HTTPS	Zone B: Our zone
Vercel: Server	Request Handler, Search Controller, Search Area Planner, Fairness Ranker, Google Connector, Server key.	Web browser: HTTPS Google Geocoding, Places and Routes APIs: HTTPS.	Zone B: Our zone
Google Maps Platform	Google Maps, directions app, Geocoding, Places and Routes APIs	Server: HTTPS. Web browser: HTTPS.	Zone C: Google

6.5 From analysis to architecture

Analysis object (from 5.1 and 5.2)	Component that owns it (6.3)	Note (split, merged, external)
Participant / start location	Search session	Keeps both participants' start addresses in memory during the search
Meeting search	Search Session, Search controller	Split between storing user choices and handling the search process.
Search Area	Search Area Planner	Calculates the midpoint and search area from both starting locations
Candidate Place	Search Controller, Results View	The controller manages candidate places, while Result View displays them. Place data comes from Google connector
Driving Time / Distance	Google Connector	Gets driving times and distances from the external Google Routes API.
Fairness Gap	Fairness Ranker	Calculates the difference between the participants' driving times or distances and ranks places
Selected Place / Directions Request	Place Actions	Handles the selected meeting place and creates a link to an external directions app.

6.6 Design principles applied

Principle	Where it shows in your design	Evidence (component, interface, ADR)
Separation of concerns	The system separates user input, search processing, ranking and map display into different components	Search Form, Search Controller, Fairness Ranker, Map View
Information hiding	Google API Keys and provider-specific details stay on the server and are not exposed to the browser.	Google Connector, LocationData, ADR-01
Low coupling	Other components use the Location Date interface instead of directly communicating with Google APIs	LocationData, Google Connector, ADR-02

Principle	Where it shows in your design	Evidence (component, interface, ADR)
High cohesion	Each component focuses on one main responsibility, such as calculating the midpoint or ranking places.	Search Area Planner, Fairness Ranker
Design for change	The location provider can be replaced without changing the rest of the application.	Google Connector, LocationData, ADR-02
Keep it simple (KISS)	The system supports only two participants and driving routes, without accounts, reservations or a database	Section 1.2, Section 6.2, ADR-01

7 Design Rationale

7.1 Architecture decision records

At least three. ADR-01 records the style decision from 6.2. Use the format you practiced in the Lesson 16 workshop, written about your own project. Copy the block for each decision.

ADR-01 Client-server: our server sits between the browser and Google

Status	Accepted
Date and owner	11/8/26, Vedanshi Sreeramoju
Context (drivers)	QAS-02 (no address may be kept), QAS-04 (only one part talks to the provider), QAS-01 (results in 5 s), CON-02 (free tier only, so the API key must not leak), UC-01, UC-02
Options considered	(1) The browser calls Google directly, with no server. (2) Client-server: the browser sends the search to our server, which calls Google. (3) Layered with our own database that saves searches and results.
Decision	Option 2. The React web app runs on the User's device, and the Request Handler, Search Controller, Search Area Planner, Fairness Ranker and Google Connector run on a Vercel server.
Consequences	Favors security and privacy (QAS-02, CON-02) and modifiability (QAS-04). The key never reaches a browser, so nobody can copy it and use up our free tier. The move is putting all provider calls on a server we control. Cost: every search takes one extra network trip, which eats into the 5-second budget of QAS-01, and the team has to build and deploy server code as well as the web app.
Evidence	None yet. Spike S-01 could not run because Google requires a $30 prepayment to activate the key (Section 8). Timing is deferred to P3.
Related	QAS-01, QAS-02, QAS-04, CON-02, R1, ADR-02, ADR-03

ADR-02 All location data goes through one Google Connector

Status	Accepted
Date and owner	11/8/26, Vedanshi Sreeramoju
Context (drivers)	CON-01 (data must come from a third party), CON-02 (free tier only), QAS-04 (switching provider changes one component), assumption A1, the P1 pricing estimate of about 250 searches a month (2.6)
Options considered	(1) Each server component calls Google wherever it needs data. (2) One Google Connector behind a LocationData interface, with OpenStreetMap (Nominatim, OSRM, Overpass) as the planned fallback. (3) Use OpenStreetMap from the start.
Decision	Option 2. Only the Google Connector talks to Google. The other components ask LocationData for locations, places and driving figures and never see which provider answered.

Consequences	Favors modifiability (QAS-04). If we run out of free usage or lose access (A1), we replace one component. The move is hiding the provider behind an interface. Cost: one more component to write, and LocationData can only offer what both Google and OpenStreetMap can supply, so we give up Google-only features. We chose Google over option 3 because its place data and drive times are more complete, at the risk of the usage limit.
Evidence	P1 economic feasibility (2.6): Route Matrix limits us to about 250 searches a month, above the about 200 we need for testing and demos.
Related	QAS-04, CON-01, CON-02, CON-03, A1, R1, ADR-01

ADR-03 Addresses live in memory only

Status	Accepted
Date and owner	11/8/26, Vedanshi Sreeramoju
Context (drivers)	QR-02 and QAS-02 (zero addresses retained after the session), CON-03 (limits on storing provider results), stakeholder need: the other participant's address is entered without them present (2.2, 2.3)
Options considered	(1) Save searches and results in a database or cache so repeat searches are faster. (2) Keep the last search in browser storage so it survives a page refresh. (3) Keep addresses only in memory for the length of the request and session, and never write them to storage, logs or URLs.
Decision	Option 3. Search Session holds the search in browser memory, and the server forgets it when the request ends. Addresses are sent in the request body, never in a URL, and server logs record no addresses.
Consequences	Favors privacy (QAS-02) and keeps us within the provider's terms (CON-03). The move is having no persistent store at all. Cost: no caching, so every search pays for its full set of provider calls, which uses free-tier quota (CON-02) and time (QAS-01). If the User refreshes the page, they retype both addresses. We also lose address logs for debugging.
Evidence	None yet. QAS-02 will be verified by inspecting browser storage, server logs and network requests after 10 sessions.
Related	QAS-02, QR-02, CON-03, ADR-01

ADR-04 A provider failure ends the search with a "try again" message

Status	Accepted
Date and owner	11/8/26, Vedanshi Sreeramoju
Context (drivers)	QAS-03 (message within 6 s, all inputs kept, no partial ranking), QAS-05 (ranking must be correct), UC-01 A4, UC-02 A2

Options considered	(1) Retry the failed provider call automatically before giving up. (2) Show a partial ranking from whichever places returned figures. (3) Stop each provider call after 4 seconds and show a “try again” message with the User’s entries kept.
Decision	Option 3. The Search Controller gives each provider call a 4-second limit, does not retry, and never ranks places with missing figures.
Consequences	Favors availability as the User sees it (QAS-03) and correctness (QAS-05). The User gets a clear answer quickly instead of a long wait or a ranking that looks fair but is built on incomplete data. The move is failing fast. Cost: a brief provider hiccup ends the search, and the User has to ask again, even when a retry would have succeeded.
Evidence	None yet. QAS-03 will be tested with 20 forced provider failures.
Related	QAS-03, QAS-05, UC-01 A4, UC-02 A2, R1

7.2 Trade-off summary

Quality favored	Architectural move	Cost accepted	Recorded in
Security, privacy	Server between the browser and Google, holding the key	Extra network trip per search; server code to build	ADR-01
Modifiability	One Google Connector behind LocationData	Extra component; no Google-only features	ADR-02
Privacy	No persistent storage; addresses in memory only	No caching, so more provider calls and quota; entries lost on refresh	ADR-03
Availability, correctness	4-second timeout, no retry, no partial results	A brief provider hiccup fails the whole search	ADR-04

8 Prototype Evidence: Architecture Spike

Spike ID and title	S-01
Risk or decision it addresses	ADR-01 / QAS-01 / QAS-03
Question (answerable yes or no)	Can one search 2 addresses, find places near the midpoint and get drive time/routes in under 5 seconds using a Google Maps API key for no cost?
Set-up	Python and Google Maps API Platform; Found at akao335/MidMeet/spikes/
How it was measured	Unable to use API key without a $30 prepayment fee.
Result (numbers)	N/A
What it changed or confirmed	Minimum cost requirement is $30. Though this payment is potentially refundable, there needs to be further discussion about the use of the API key and feasibility of project before I tie my debit card information to the account.
Where to see it	akao335/MidMeet/spikes/

9 Traceability

9.1 Extended traceability matrix

Requirement	Use case	Analysis classes	Component(s)	QAS / ADR	Evidence
FR-01	UC-01	Participant, StartLocation	Search Form, Search Session, Google Connector	QAS-02, ADR-03	Acceptance test in 3.1; planned for P3
FR-02	UC-01 (A1)	StartLocation	Search Form, Request Handler, Google Connector	—	Acceptance test in 3.1; planned for P3
FR-03	UC-01, UC-02	MeetingSearch, MeetingOption	Search Form, Search Session, Fairness Ranker	QAS-05	10-pair miles/minutes test (3.1)
FR-04	UC-01, UC-02	MeetingOption, Place	Search Controller, Search Area Planner, Google Connector, Results View	QAS-05, ADR-02	10-pair result-count test (3.1)

9.2 Gap analysis

Gap	Items	What you did about it
Requirements with no component	FR-10, FR-11, CON-04	FR-10 and FR-11 are WON'T items, so having no component is intended. CON-04 is a process constraint met by the repo and milestones, not by runtime code. QR-03 has components but no QAS or ADR; it will be verified by the usability session in P3.
Components with no requirement, QAS or ADR	Place Actions (share message, UC-04)	The sharing behavior has no FR. Propose adding FR-12 (COULD: share the chosen place) as a 0.2 change in P3, or drop sharing from Place Actions.
Use cases not analyzed (and why that is acceptable)	UC-04 Share the chosen place	UC-04 is an optional extension of UC-02. It only formats data already on screen, adds no provider calls, and carries no requirements yet, so it has no non-trivial steps to analyze.

10 Project Plan (updated; was P1 section 4)

10.1 Roles and working agreement

10.2 Plan to P3

Milestone	Date	Owner	Done when
Further explore use of Google Maps API key and create stricter restrictions on API key use in order to ensure zero cost use.	11/8/26	Nick Jackson	11/15/26
Draft wireframes for the main screens and checks each against the functional requirements it shows	11/8/26	Alison Kao	11/15/26
Extend the traceability matrix to P3 design classes and test cases, and revisit ADR-02 now that the spike found Google needs a $30 prepayment	11/8/26	Vedanshi Sreeramoju	11/15/26

10.3 Risks (updated)

Risk	Status since P1	Likelihood / impact	Mitigation and owner
R1			
R2			
R6 (new)			

10.4 Repository

10.5 Use of AI tools in P2

Member	Tool	What it was used for	What you checked or changed
Nicholas Jackson	Claude	Developing spike S-01	Checked functionality of generated code and ensured proper use of API key.

Member	Tool	What it was used for	What you checked or changed
Yair Parkhov	Claude	Developing the 6.3 and 6.4 diagrams	Checked the diagrams and rewrote the draft
Vedanshi Sreeramoju	Claude	Drafting ADR-01-04, the 7.2 trade-off table, and the 3.4 and 3.5 text	Checked each ADR against 6.1–6.4 and the QAS, and edited the diagram myself in draw.io
Sameeraa Kandalgaonkar	Claude	Developing and referencing for sections 9.1 and 9.2	Checked and reviewed details against the document.

11 Open Questions (updated; was P1 section 5)

ID	Question	Raised by	Owner and due date	Status or answer	Assumption meanwhile
Q3	What should happen if no meeting places are found within the selected category or maximum driving limit?	Nicholas Jackson	Nicholas Jackson (11/8/26)	The platform will notify that user that there are not any meeting places that meets the specified requirements and will prompt user to expand criteria.	There is not a combination of addresses that will not have a meeting place with any given specifications.
Q					

12 Baseline v1.0 Declaration (was P1 section 6)

We declare this document, version 1.0, the baseline for P2. Sections 1 to 9 describe what we will build in P3. Any later change to them is recorded in section 0.2 with its reason.

Team member	Signature or initials	Date
Nicholas Jackson		11/08/26
Yair Parkhov	Yair Parkhov	11/08/2026

Team member	Signature or initials	Date
Alison Kao	Alison Kao	11/08/26
Tanay Munnangi	Tanay Munnangi	11/08/26
Vedanshi Sreeramoju	Vedanshi Sreeramoju	11/08/26
Nandhana Kamalabaskaran	Nandhana Kamalabaskaran	11/08/26
Sameeraa Kandalgaonkar	Sameeraa Kandalgaonkar	11/09/26